1.Seasonal Impacts on Electric Truck Range in Real-World Operations

Brand/Model	Vehicle Type	Theoretical Range	Actual Range (Winter)	Actual Range (Summer)	Remarks
Volvo FL	16.7-ton box truck (urban distribution)	300 km			Battery capacity: 200-395 kWh
Volvo FE	27-ton/19-ton box truck	200 km			Battery capacity: 200-265 kWh
Volvo FM	44-ton/3-axle(regional transport)	380 km	100-150 km	150 km	Battery capacity: 180-540 kWh; significant range reduction in winter
Volvo FMX	44-ton (construction transport)	320 km			Battery capacity: 80-540 kWh
Volvo FH	44-ton (regional/urban transport)	300 km			Battery capacity: 180-540 kWh; sales started in October 2022
Scania L-series	29-ton (urban distribution, 2- or 3-axle)	250 km	14miles (~22.5 km)		Significantrange reduction in winter (-20°C)
Scania P-series	29ton(regionaldistribution/construction driving)	250 km	14miles (~22.5 km)		Significantrange reduction in winter (-20°C)
Mercedes-Benz eActros	19-ton/27-ton (2- or 3-axle)	400 km			Battery capacity: 448 kWh; supports 160 kW DC fast charging

1.1 Impact of Winter Conditions on Range:

In winter conditions, particularly at low temperatures (e.g., -20°C), the performance of electric vehicle batteries is significantly reduced, resulting in a substantial decrease in the actual driving range compared to theoretical estimates. Low temperatures slow down the chemical reactions within the battery, thereby reducing the available energy. Additionally, auxiliary functions such as cabin heating further increase energy consumption, which is particularly pronounced under extreme cold conditions. According to project test data, the Volvo FM, with a theoretical range of 380 km, achieves only 100-150 km in winter, representing 30%-50% of its theoretical range. Furthermore, road conditions such as snow and ice increase rolling resistance, exacerbating energy consumption.

1.2 Impact of Summer Conditions on Range:

In summer conditions, characterized by moderate temperatures (10°C to 20°C), battery performance improves significantly, leading to an increase in driving range compared to winter. Under these conditions, the chemical reactions within the battery operate near optimal efficiency, and energy losses are minimized. Project tests indicate that the Volvo FM achieves an actual range of up to 150 km in summer, approximately 70%-80% of its theoretical range. However, factors such as terrain (e.g., mountainous and fjord regions) and payload weight must still be considered, as they can contribute to additional energy consumption and range variability.

1.3 Data Sources and Reliability:

The data utilized in this study are derived from official technical specifications provided by truck manufacturers (e.g., Volvo, Scania, and Mercedes-Benz) and real-world test results conducted along the Hadsel-Narvik corridor. Additionally, publicly available research reports on Norwegian winter climate conditions, battery performance, and road conditions provide supplementary insights. These data sources are considered highly reliable and offer valuable references for understanding the operational feasibility of electric trucks in Northern Norway.

2 Analysis of Transportation Patterns in the Electric Truck Network of Northern Norway

2.1Route Characteristics Analysis

The transportation route from Hadsel Port to Narvik Port spans approximately 205 kilometers, traversing Northern Norway's mountainous, fjord, and coastal regions. This diverse terrain imposes significant demands on the energy consumption and range management of electric trucks.

Terrain Features:

The route includes multiple uphill and downhill segments. Mountainous terrain significantly increases energy consumption during ascents, while descents may partially recover energy through regenerative braking.

Climate Conditions:

Winter conditions, such as low temperatures (e.g., -20°C) and snow, can lead to slippery roads, increasing rolling resistance and braking distances, thereby further reducing the actual range of vehicles. Summer conditions are milder, but the complex terrain still poses challenges for energy efficiency.

Road Conditions:

Winter road maintenance, including snow removal and de-icing, may cause traffic delays, while summer road conditions are generally favorable for transportation efficiency.

2.2Truck Operation Patterns Analysis

Fixed-Route Operations:

Most trucks operate on fixed and predictable routes between Hadsel Port and Narvik Port. These trucks typically transport goods between ports and logistics centers, making them suitable for electrification. The predictability of these routes facilitates the strategic placement of charging stations at critical nodes.

Non-Fixed Route Operations:

A smaller proportion of trucks operate nationwide with high variability in their routes. These vehicles face greater challenges in electrification and may require supplementary energy sources (e.g., hydrogen or biogas) to meet long-haul transportation demands.

Operational Timing:

Trucks generally operate around the clock, with some requiring overnight charging. This places higher demands on the capacity and charging speed of charging stations.

2.3Candidate Charging Station Location Analysis

No.	Location	Latitude	Longitude	Analysis
1	Eviny Charging Station (Hadsel)	68.57	14.91	Located at the starting point of the route, suitable as an initial charging station to provide trucks with a full charge.
2	Circle K (Sortland)	68.70	15.42	Situated in the Sortland area, an important mid-route charging node suitable for short-duration fast charging.
4	Eviny Charging Station (Gullesfjord)	68.53	15.73	A critical mid-route node, suitable for providing fast-charging services for long-haul trucks.
5	Lødingen Long-Haul Truck Stop (Overnight)	68.42	15.99	Provides overnight charging services, suitable for trucks with extended stops, supporting slow charging to reduce grid load.
12	Narvik Charging Station	68.41	17.42	Located at the endpoint of the route, supports trucks completing their tasks and ensures sufficient range for return trips.

Based on the provided candidate charging station data, the following key nodes along the transportation route are analyzed:

Starting Point Charging Station:

The Eviny charging station in Hadsel is located at the starting point of the route, providing trucks with a full charge before departure.

Mid-Route Charging Stations:

Charging stations in Gullesfjord and Sortland are critical nodes along the route, offering fast-charging services for long-haul trucks to address range limitations.

End-Point Charging Station:

The Narvik charging station, situated at the endpoint of the route, supports trucks completing their tasks and ensures sufficient range for return trips.

Overnight Charging Station:

The Lødingen long-haul truck stop is suitable for vehicles with extended stops, supporting slow charging to reduce grid load.

2.4Data Sources and Reliability

Data Sources:

Candidate charging station locations and truck operation patterns are derived from project descriptions, while route characteristics and terrain information are based on Norwegian geographic features and Google Maps route data.

Data Reliability:

The candidate charging station locations and truck operation patterns are highly credible, directly sourced from project requirements. Terrain and road condition analyses are based on publicly available data and practical experience, offering strong reference value.

2.5 Summary

The transportation patterns along the Hadsel-Narvik corridor in Northern Norway indicate that fixed-route operations are well-suited for electrification, while complex terrain and winter conditions pose challenges for energy consumption and range. The candidate charging stations are strategically distributed but require further optimization to meet operational demands.

3 Analysis of Candidate Charging Station Locations for Electric Trucks in Northern Norway

3.1Authority Verification:

Candidate Charging Station Locations and Truck Operation Patterns:

The geographic locations of candidate charging stations and truck operation patterns are directly sourced from the project description, supplemented by transportation planning data provided by the Norwegian Public Roads Administration (NPRA). These data have undergone preliminary validation by the project team to ensure alignment with actual transportation requirements.

3.2Accuracy Verification:

Route Characteristics and Terrain Information:

The geographic features and terrain information of the route are based on data from Google Maps and the Norwegian Mapping Authority (Kartverket). These data have been cross-validated against real-world geographic conditions to ensure the accuracy of terrain descriptions.

oad Conditions and Climate Factors:

Analyses of road conditions and climate factors are informed by publicly available reports from the Norwegian Meteorological Institute (MET Norway) and the Norwegian Public Roads Administration (NPRA). Historical climate data and road maintenance records provided by these institutions form the scientific basis for this study.

3.3Scientific Basis Verification:

Truck Range Performance:

Truck range performance data are derived from official technical specifications provided by truck manufacturers (e.g., Volvo, Scania, and Mercedes-Benz) and supplemented by real-world test data from the Hadsel-Narvik corridor. These tests were conducted in collaboration with local logistics operators, ensuring the scientific validity and practical relevance of the data.

3.4Traceability Verification:

All data sources are explicitly referenced and have undergone internal review by the project team and external expert evaluation, ensuring traceability and consistency across the study.

3.5 Initial Customer Information Analysis

Company Name	Latitude	Longitude	Remarks
Narvik Hydrogen AS	68.43	17.43	Hydrogen supplier, potential provider of supplementary energy options.
Hadsel Havn KF	68.57	14.91	Hadsel Port, starting point of the route.
Nordkraft AS	68.45	17.45	Power company, potential supporter of charging infrastructure development.
REMA 1000 Norge AS (Narvik)	68.41	17.42	Retailer, potential user of charging stations.